

VALESKA JUDITH MOLINA MOLINA

PhD Candidate in Astronomy & Planetary Science | Astrobiology

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EDUCATION

- PhD Candidate in Astronomy and Planetary Science, Universidad de Atacama (2023–Present)
- MSc in Astronomy (Astrobiology), Universidad de Antofagasta (2022)
Thesis: Atmospheric modelling of rocky exoplanets using CO spectra produced by *Roseovarius* sp.
- BSc & Professional Degree in Biochemistry, Pontificia Universidad Católica de Chile (2015–2016)
- Diploma in Science Communication, Universidad de Antofagasta (2021)

RESEARCH INTERESTS

Astrobiology, biosignatures, exoplanet atmospheres, microbial metabolism, spectroscopy (Raman/IR), extreme environments

PUBLICATIONS

P Longa-Peña, J Tregloan-Reed, C Adam, J Chamoun-Contreras, E Unda-Sanzana, C Ávalos-Vega, F Amadio, M Andersen, M Bonavita, V Bozza, B Campos Estrada, M Dominik, A Donaldson, R Figuera Jaimes, J Fynbo, T C Hinse, M Hundertmark, U G Jørgensen, E Khalouei, M Kretlow, **V Molina**, N Peixinho, M Rabus, S Rahvar, A R Rajkumar, M I Romero-Colmenares, P Rota, S Sajadian, J Skottfelt, C Snodgrass, J Southworth, H Alarcon, G Blanc, P Bocaz, J P Colque, J Cortes, C Flores Quintana, P Garcia, R González, S Kim, S Martín, T Nakos, E Ortiz, A Otarola, J Plaza Hernández, P Sanhueza, G Siringo, M Soto, Mindstep collaboration:, CLEOsat collaboration:, Brightness evolution of LEO Starlink mega-constellation satellites from 2021 to 2023: a multiyear ground-based photometric study, *Monthly Notices of the Royal Astronomical Society*, Volume 548, Issue 2, May 2026, [stag552](#).

Molina V. et al. (2026). Exploring extremophile gas production as a biomarker for early Earth atmospheres. *International Journal of Astrobiology*.

Tregloan-Reed J., **Molina V.** et al. (2021). Optical-to-NIR magnitude measurements of Starlink Darksat. *A&A*, 647.

Tregloan-Reed J., **Molina V.** et al. (2020). First observations of Starlink Darksat. *A&A*, 637.

Book: *Comunicándonos: Secretos de las ciencias extremas* (2022) ISBN 978-956-8293-84-0

PROFESSIONAL & ACADEMIC EXPERIENCE

- **Regional Ministerial Secretary of Science** (SEREMI CTCI), Chile (2025–2026)

Government-appointed regional authority responsible for implementing national science policy. Led strategic initiatives in gender equity, scientific decentralization, and public engagement, fostering collaboration between academia, government, and local communities.

- **Assistant Professor, Universidad del Alba** (2022–2023)

Teaching in biochemistry, biology, histology, chemistry, and curriculum development.

- **Research Assistant / Technical Staff**, Universidad de Antofagasta (2019–Present)

- **MSc Researcher**, Universidad de Antofagasta (2019–2022)

Cultivation of *Roseovarius* sp. and Raman/IR spectroscopy for exoplanet analogues.

- Additional roles in teaching, molecular biology research, and clinical sampling (2014–2021).

RESEARCH PROJECTS

Co-PI, ESO–Chile Indigenous Knowledge & Astronomy Project (2024)

Researcher, Extreme Science Outreach Center (ANID, 2021–2024)

Researcher, Takkini: From the desert to the universe (2021–2024)

Technical Staff, Microbial Complexity Projects (2019–2021)

PI, Red abalone cell immortalization (2016–2019)

CONFERENCES & PRESENTATIONS (SELECTED)

ISME 19 (2024), IAU General Assembly (2024), Arizona Astrobiology Symposium (2024)

SOCHIAS annual meeting (2024), Origins Conference (2023), Latin American Astrobiology Congress (2021). RED Astrobiology course (2021).

Multiple international and national presentations in astrobiology and microbiology.

FUNDING/GRANTS

2025 — Gastos Operacionales, ANID, Chile.

2024 — Travel Grant to attend the IAU General Assembly 2024, Cape Town.

2024 — Travel Grant to attend the ISME 19 2024, Cape Town.

2023 — Travel Grant to attend the Origins 2023, Quito.

2023 — National doctoral scholarship (competitive), Agencia Nacional de Investigación y Desarrollo, Chile.

AWARDS & FELLOWSHIPS

Finalist, L'Oréal For Women in Science (2024)

40 Leaders of Atacama (2024)

SCIENCE COMMUNICATION

Science outreach, educational content creation, public talks, and workshops in astrobiology.

OBSERVING EXPERIENCE

La Silla Observatory (ESO): 27 nights (2024–2025)

Ckairama Observatory: 10 nights (2019–2021)